

AI & Filmmaking: A Sandbox for Generative AI Experimentation in Computer Graphics Education

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1 GENERATIVE AI IN FILM MAKING

Generative Artificial Intelligence (AI) is rapidly reshaping creative industries, including filmmaking. In education, the integration of generative AI presents both opportunities and challenges, requiring new pedagogical approaches that allow students and faculty to experiment, adapt, and critically engage with these emerging tools in a supportive environment. This paper examines a case study of a course offered in Fall 2024 called “*AI & Filmmaking: A Critical Exploration*”, taught by Anthony Giacchino, which was designed as a sandbox environment within the MFA Computer Arts program at the School of Visual Arts. Students collaborated on the production of a silent documentary short film centered on a historical event, employing generative AI tools alongside traditional animation and filmmaking techniques. Faculty-led research on Joe Petrosino, a notable early 20th-century New York detective and Italian immigrant, provided the factual foundation for the project. One of the learning outcomes was to critically examine how AI-generated visuals can reenact historical events while addressing the genre’s demand for authenticity. This talk aims to analyze the learning outcomes, artistic and technical limitations, and ethical considerations. By positioning AI as both a creative collaborator and a technical tool, the course provided a structured yet flexible framework for when

the tools evolve or change. It followed a standard documentary filmmaking process: researching, field trips, writing the script, art direction, “filming”, editing, closed captioning, visual effects, sound design, and credits. And the use of generative AI: ChatGPT for scripting, translating documents, and writing prompts, Midjourney for generating images, and Runway to generate videos and lip syncing. The final stage involved editing and post-production, where a student from the VFX team applied approximately 120 effects, including film grain, scratches, and era-specific textures, to achieve the authentic look of a 1909 silent film.

2 SANDBOX AS AN EXPERIENTIAL LEARNING MODEL

The course operated as a sandbox, a pedagogical model that fosters exploration, iterative learning, and hands-on experimentation without rigid constraints. By creating a flexible framework where AI tools could be freely tested and substituted as they evolve, students may engage in an iterative production cycle, refining their techniques. AI-assisted scriptwriting facilitated the development of narrative structures, frame interpolation automated labor-intensive tasks, and fast experimentation with aesthetics. However, students encountered various technical limitations, such as inconsistencies in AI-generated outputs, difficulties in achieving a coherent aesthetic across different AI-generated assets. We found that the shift from passive consumption to active engagement helped to develop a sense of agency in their creative work, equipping them with the adaptability needed to work with rapidly evolving AI technologies, which are foreseen in professional settings. By May of 2025, these limitations will have drastically diminished within most AI video generation tools, such as Runway, Luma, and Google Veo. The sandbox approach was considered beneficial since the structure of the class can remain intact, and only the AI tools, which evolve and improve at a fast pace, can be exchanged in the film-making pipeline.

3 ETHICAL CONSIDERATIONS IN AI-ASSISTED FILMMAKING

The integration of generative AI into documentary filmmaking raises complex ethical questions surrounding truth, authorship, and historical representation. While AI-generated imagery can create visually compelling historical reenactments, it also presents significant concerns regarding accuracy, bias, and manipulation; we believe in a balance between authenticity and fabrication. Another major ethical concern is the potential bias inherent in AI training data. AI models are trained on existing datasets, which may

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Figure 1: Joe Petrosino during a “seat-down” interview. Film still from lip sync video generation in Runway. Image by Anthony Giacchino and his Fall 2024 students.

reinforce historical inaccuracies or cultural biases present in the source material. For that, we instituted careful curation and oversight. Transparency and attribution are also crucial considerations in AI-assisted filmmaking. The ethical use of AI in documentary production necessitates clear disclosure of AI-generated content, ensuring that audiences are aware of the extent to which AI has been used in the filmmaking process. Finally, the integrity of documentary filmmaking as a truth-seeking art form is at stake. In September 2024, the Archival Producers Alliance released the best practice guide for the use of AI in documentary filmmaking.

4 DISCUSSION: AI’S ROLE IN REENACTING HISTORICAL EVENTS

One of the most compelling aspects of the project was the ability to reenact historical events through visual storytelling. It may provide

unique opportunities for creative exploration and education, but it also introduces a range of challenges related to historical accuracy, stylistic consistency, and narrative impact. Ensuring the accuracy of AI-generated historical visuals requires rigorous fact-checking against historical records. Students needed to evaluate the credibility of AI-generated depictions and cross-reference them with primary sources, such as historical images from the New York Public Library or the Library of Congress, to avoid inaccuracies. The project demonstrated that while AI can enhance historical storytelling, human oversight remains essential in shaping compelling and responsible narratives. Looking ahead, future iterations of the course will deepen the focus on ethical implications at every stage of the creative process, contributing to broader discussions on AI’s evolving role in media arts education and its impact on future creative industries.

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